Workshop No. 2
Design Artifacts and System Modeling
Part 1

Project: MusicLock

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1. System domain

MusicLock is a web application for storing, playing, and distributing music focused on user privacy. Traditional audio storage and streaming platforms have direct access to their users' files on their servers. This exposes them to data leaks and unauthorized access by third parties or the administrators themselves.

MusicLock resolves this problem with Client-Side Encryption. MP3 files are encrypted in the user's browser using the Web Crypto API and the AES-GCM algorithm before being sent and stored in the cloud. Each user has a master key generated securely and protected with a key derived from their password via Argon2id. As a result, neither the server, the database, nor the object storage service has access to the keys required to decrypt the content. Decryption happens transparently and exclusively in the browser when playing or downloading a file.

The only files that are not encrypted with user-specific keys are those published as public. For these files, MusicLock acts as a standard media distribution service: unencrypted streams are processed locally or via public keys to enable open access, and MusicLock integrates the third-party ACRCloud API to scan the unencrypted audio fingerprint for copyright compliance prior to distribution.

In addition to private storage, the platform offers social and music organization features:

- Album publishing
- Creation of private or public playlists.
- User following.
- Comments on songs and public playlists.
- Favorite songs and playback history.

The system operates on a two-tier subscription model, Free(standard) and Premium, which defines the storage, upload, download and sharing limits for each account.

System Boundaries:

- **Inside the system:** The web application (client), the application server and the database.
- **Outside the system:** The ACRCloud copyright identification service.

1.1. Describe the actor's characterization.

Actor 1: Standard User (free users)

Attribute	Description
Type	Human actor, primary

Description	Registered person that uses the platform with the free plan.
Objectives	Store their music privately and securely, play it from any device, organize it, and share it with other users.
Functionalities	Register and log in; encrypt, upload, play, and download MP3 files; publish albums; create playlists; mark favorites; view their playback history; follow other users; comment on public content; share files and playlists; view plan comparisons and upgrade their subscription.
Restrictions	Maximum storage of 500 MB; maximum file size of 20 MB; maximum 10 file downloads per month and a limit of 5 persons to share each file or playlist in private mode.

Actor 2: Premium User

Attribute	Description
Type	Human actor, primary (specialization of the standard user)
Description	Registered person with an active Premium subscription. Has all the functionalities of the Standard User, with expanded limits and additional benefits.
Objectives	The same as the Standard User, plus managing a larger volume of music, organizing it across the playlist, and gaining insights into their listening habits.
Functionalities	Access advanced listening statistics (total listening time, top artists, and top tracks over the last 7 and 30 days); unlimited downloads; sharing without user limits; view subscription status and downgrade plan.
Restrictions	Maximum storage of 5 GB; maximum file size of 100 MB; benefits are dependent on an active subscription.

Actor 3: System Administrator

Attribute	Description
Type	Human actor, support
Description	Person in charge of the operation and supervision of the platform.
Objectives	Maintain the proper functioning of the system and manage accounts and plans.

Functionalities	Block or suspend user accounts; manage Premium plans; review subscription statuses; monitor server performance.
Restrictions	Cannot access the content of user files, as they do not possess the decryption keys.

Actor 4: ACRCloud

Attribute	Description
Type	External system
Description	Audio identification service that checks for potential copyright infringements in files that users wish to make public.
Objectives	The system sends an audio sample via its Identification API (https://docs.acrcloud.com/reference/identification-api/identification-api) and receives the identification results.
Functionalities	Verify if the file meets the copyright policy.
Restrictions	Only processes files designated for public release, never private files.

Actor 5: Non-Registered User

Attribute	Description
Type	Human actor, primary
Description	An anonymous user who accesses the web without logging in.
Objectives	Get the web main web info: its functionalities, the current existing plans and the group and project summary.
Functionalities	Can create a new account or login into a registered one.
Restrictions	Can’t upload or listen public or private files on the web.

1.2. Present the business rules.

Accounts and Security

- **BR-01:** Every user must register and log in to access the platform.
- **BR-02:** Private files must be encrypted in the user's browser (Client-Side) using AES-GCM before being sent to the server. The server never receives or stores private files in plain text.
- **BR-03:** Each user's master key is derived from their password using Argon2id. Neither the server nor administrators have access to it.
- **BR-04:** Decryption of private content occurs exclusively in the user's browser during playback or download.
- **BR-05:** In case of a forgotten password, account and file recovery requires a Recovery Key (Seed Phrase) generated during registration. Without this key, encrypted files cannot be recovered.
- **BR-06:** A blocked or suspended user cannot log in or access the platform API.

Plans and Limits

- **BR-07:** There are two plans: Free (Standard) and Premium. All new registered users start by default on the Free plan.
- **BR-08:** Total storage limit is 500 MB for the Free plan and 5 GB for the Premium plan.
- **BR-09:** The system validates available storage before allowing any upload; if the user exceeds their quota, the upload is rejected.
- **BR-10:** Maximum file size is 20 MB for the Free plan and 100 MB for the Premium plan.
- **BR-11:** In the current version, the system only accepts audio files in MP3 format.
- **BR-12:** Free plan users can perform up to 10 original MP3 file downloads per month. Downloads are unlimited on the Premium plan.
- **BR-13:** Content sharing on the Free plan is limited to a maximum of 5 users per file or playlist in private mode. The Premium plan has no sharing limits.
- **BR-14:** Access to advanced listening statistics is exclusive to Premium accounts.

Subscriptions and Account Status

- **BR-15:** Users can change their plan (upgrade or downgrade) at any time.
- **BR-16:** The system notifies the user 7 days prior to the expiration of their Premium subscription.
- **BR-17:** If a Premium subscription expires and is not renewed, the account automatically reverts to the Free plan.
- **BR-18:** If an account exceeds the 500 MB limit upon reverting to Free, it enters a "Frozen" state: existing files remain intact and playable, but new uploads are blocked.

The user has a 90-day grace period to renew Premium or delete files before excess data is permanently purged according to the upload date.

Content, Publishing, and Social Interaction

- **BR-19:** All uploaded files are private by default.
- **BR-20:** To publish a public file or playlist, the client browser must generate an audio fingerprint and send it to ACRCloud. If a copyright violation is detected, the content cannot be made public.
- **BR-21:** To share encrypted files with specific users, the system uses an asymmetric key exchange (Public/Private Keys) without exposing the users' master keys.
- **BR-22:** Comments and interactions are permitted exclusively on songs and playlists marked as public.
- **BR-23:** A public file cannot be converted into a private file, the only way to do it is by deleting the public one, and re-uploading it but with private settings.

2. CRC Cards

2.1. Main Classes (Identify & Describe)

The classes used for the system are:

- **User:** Represents any person registered on the platform (listener or creator), controlling their profile, available storage, and follow relationships.
- **Album:** Is the entity responsible for modeling and managing official musical groupings within the platform.
- **Song:** Models the system's multimedia component, containing the audio files' metadata, copyright status, and encrypted location.
- **Playlist:** Defines personalized collections of songs that can be private, public, or shared with collaborative roles among multiple users.
- **Comment:** Manages feedback text and interactions that users post within specific songs or playlists.

2.2. CRC Cards (Class-Responsibility-Collaborator)

Card 1: User Class

Class: User	
Description: Represents any account registered in the system. Controls both the features of a standard listener (profile, subscription, playlists) and those of a creator/artist (uploading albums).	
Responsibilities (What it does / knows)	**Collaborators** (Interactions)
● Know access credentials (Email and Password). ● Manage subscription tier (Free/Premium) and control disk space (Max_Storage, Used_Storage). ● Manage contact networks (know which other users they follow or who follows them). ● Publish complete recording projects on the platform (Albums).	• User (for the followers feature) • Album • Playlist • Song (Favorites and history)

Card 2: Album Class

Class: Album	
Description: Official musical collection uploaded by a creator user.	
Responsibilities (What it does / knows)	**Collaborators** (Interactions)
● Store its title and release date. ● Identify the creator of the work through the relationship with the user. ● Group and order a set of music tracks (Songs).	• User (Owner) • Song

Card 3: Song Class

Class: Song	
Description: Models the system's audio multimedia files, their basic metadata, and their accessibility.	
Responsibilities (What it does / knows)	**Collaborators** (Interactions)
• Store technical information and metadata (id_song, Title, duration, size_bytes). • Locate and play the audio file using the encrypted_file_path property. • Validate distribution permissions (Is_public, Copyright).	• Album • User (To keep track of his history) • Playlist

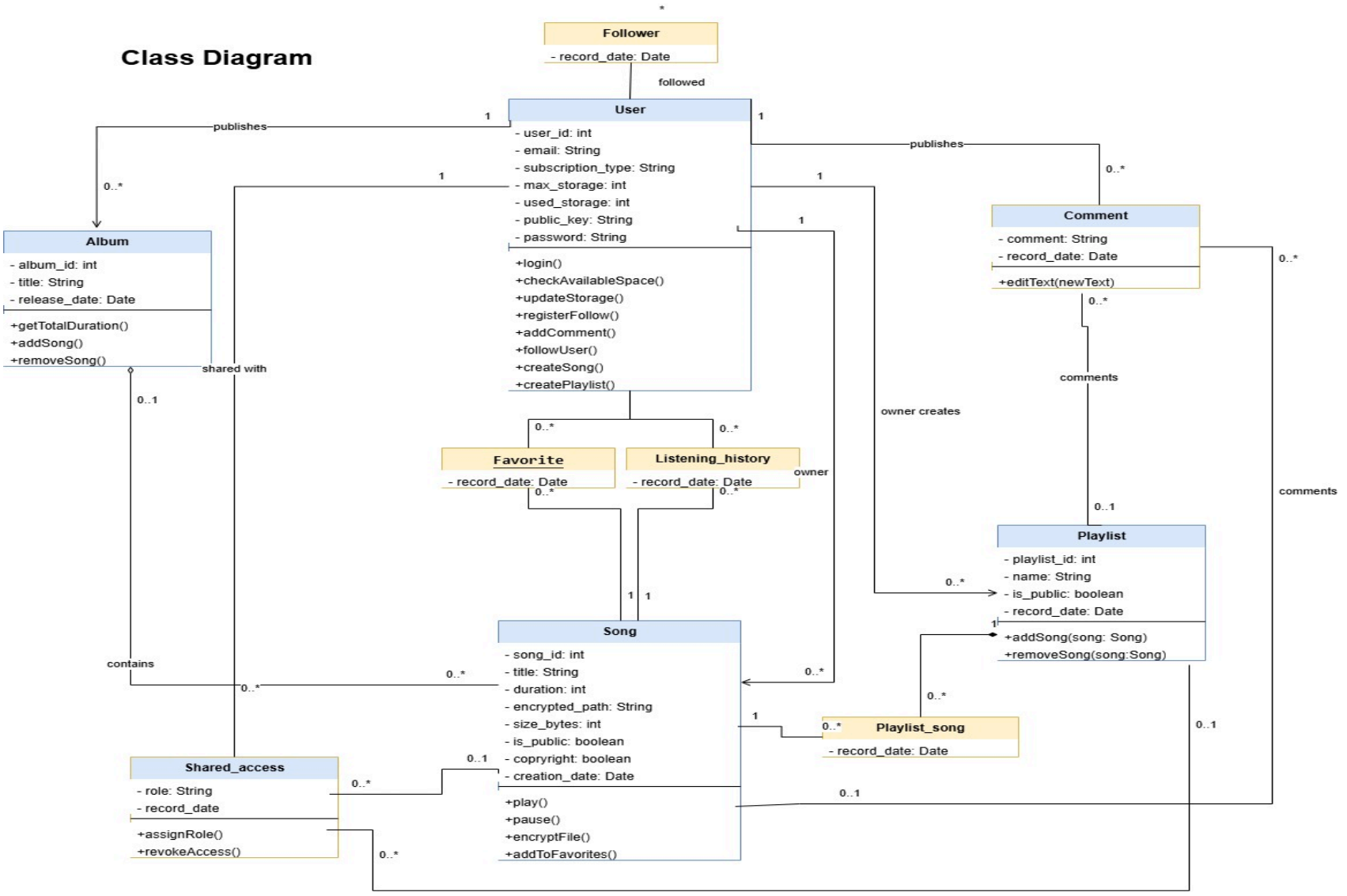
Card 4: Playlist Class

Class: Playlist	
Description: Custom or public collections of songs created by users or collaboratively.	
Responsibilities (What it does / knows)	**Collaborators** (Interactions)
• Maintain an ordered list of linked tracks. • Modify the playlist's privacy settings (is_public). • Manage editing permissions for other collaborating users (shared_access).	• User (Creator/Collaborator) • Song

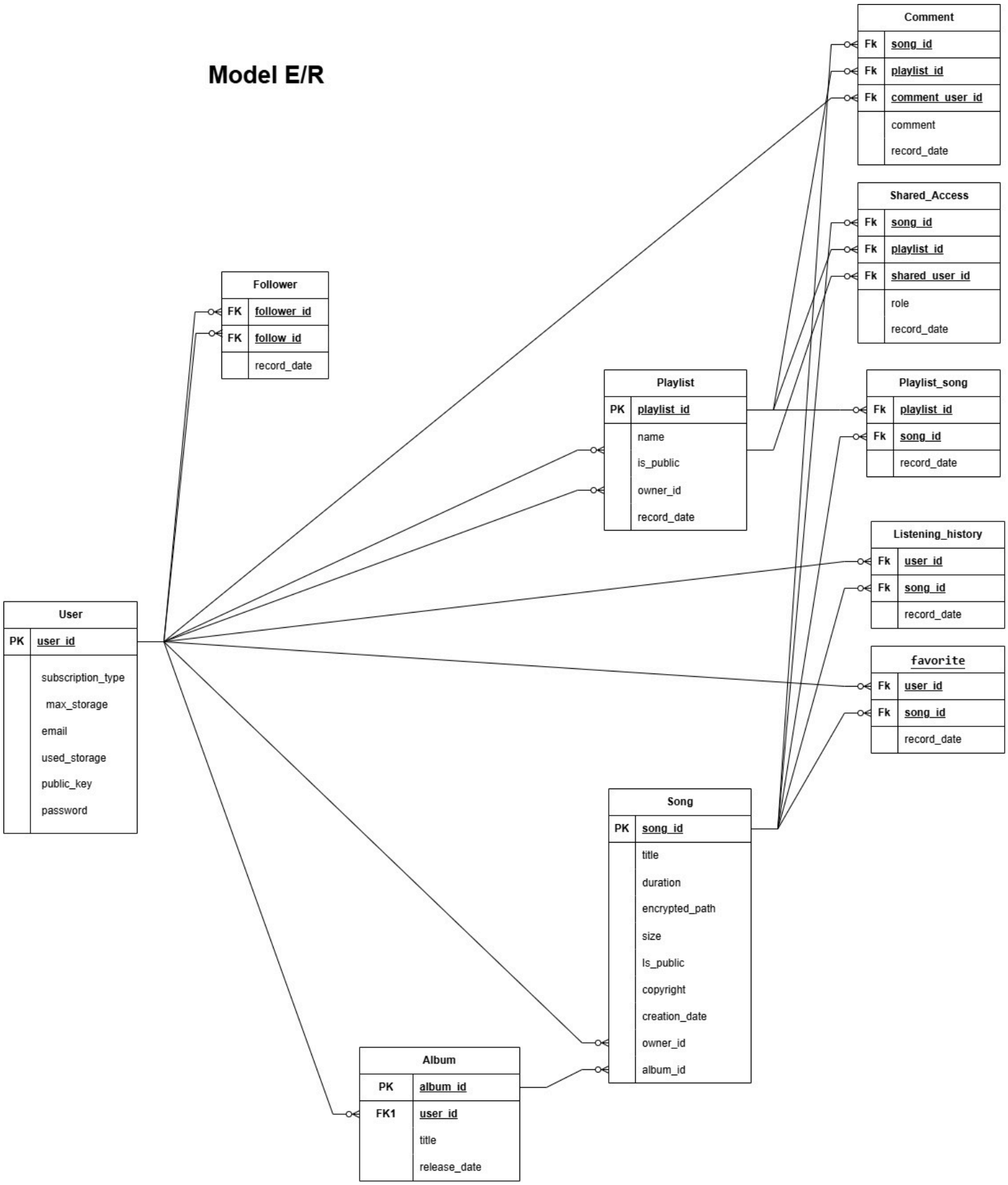
Card 5: Comment Class

Class: Comment	
Description: Represents the text interactions and feedback that users leave on system elements, such as songs or playlists.	
Responsibilities (What it does / knows)	**Collaborators** (Interactions)
● Store the written message content (Text) and its publication timestamp (registration_date). ● Identify the message author and link directly to a Song or a Playlist.	• User • Song • Playlist

3. Class Diagram



4. Relational Database Model



5. Mockups • Create mockups or wireframes for your screens.

This document outlines the visual identity, primary interfaces, and system flow for **MusicLock**, a client-side encrypted zero-knowledge audio platform. **Stitch** was utilized during the mockup phase to connect screens and simulate the functional user experience.

5.1. Color Palette

The interface utilizes a sophisticated, dark-themed hierarchy engineered for high-contrast audio production environments.

- **Primary Palette (#025098):** Deep cobalt blue used for primary branding, highlighted active storage states, and global identity blocks.
- **Secondary Palette (#849885 / Deep Teal Accent):** Applied to active playback states, successful connectivity status, and prominent UI nodes.
- **Tertiary Palette (#885CF6):** Royal violet used to represent secure keys, encrypted hashes, and metadata separation.
- **Neutral Palette (#990016):** Vibrant accent red for system constraints, restricted elements, or severe structural notifications.



Figure 1. Color Palette

5.2. Landing Page (Main Screen)

Introduces the core value proposition of MusicLock—secure, high-fidelity cloud storage completely guarded against third-party data tracking or eavesdropping.



Figure 2. Landing Page First View

The **Main Screen** features interactive navigation buttons that grant users direct access to core platform sections, including **'How It Works'**, **'Benefits'**, **'Plans'**, **'FAQs'**, and the secure **'Login'** portal.

How it works?

Explains specialized cryptographic steps using terminology accessible to music producers and sound engineers.



Figure 3. How it works? Section

Benefits

Highlights core competitive advantages against traditional, commercial storage solutions



Figure 4. Benefits section

Plans

Transparent price breakdowns for varying tiers of storage capacity.

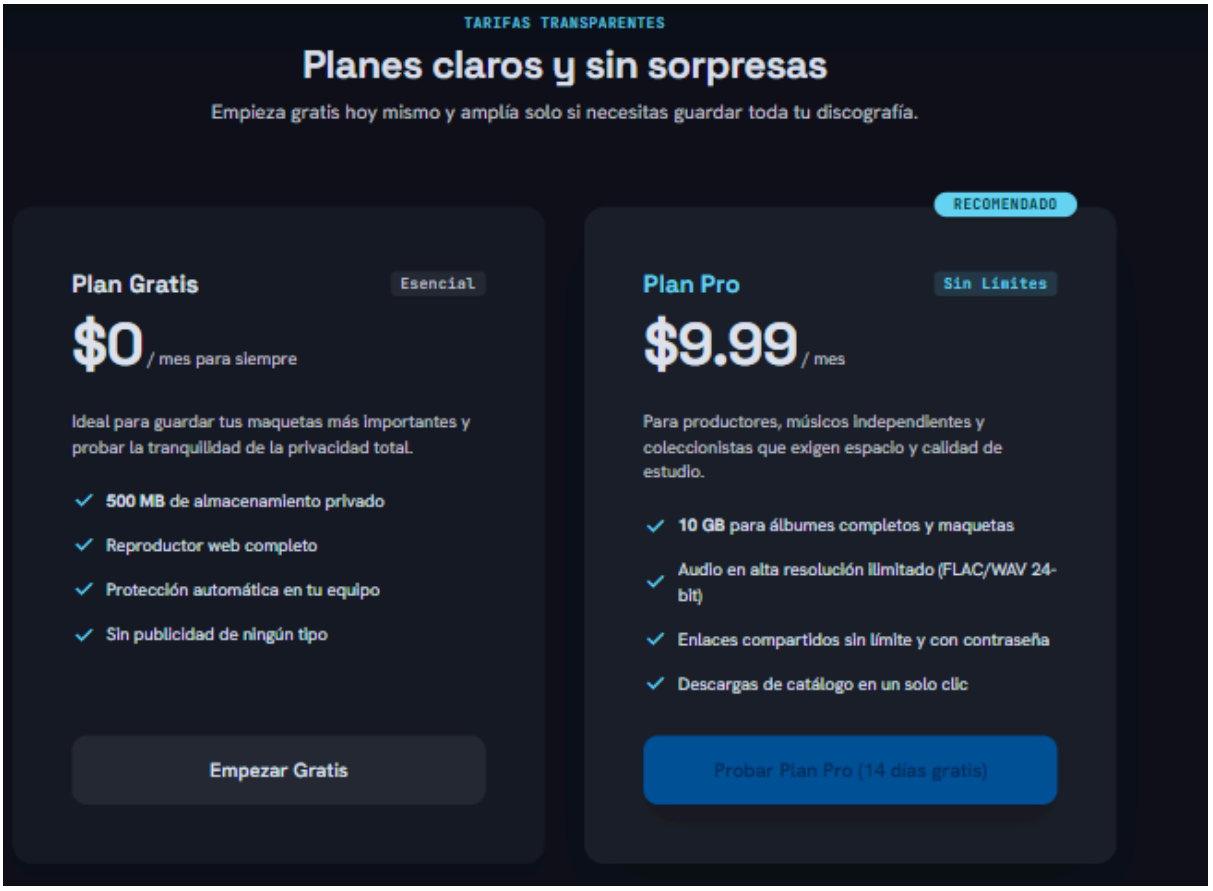


Figure 5. Plans and subscriptions section

Frequently Asked Questions

Resolves technical concerns regarding operational access and data autonomy.



Figure 6. Frequently Asked Questions Section

Login Screen

The entry boundary into individual user encryption spaces.

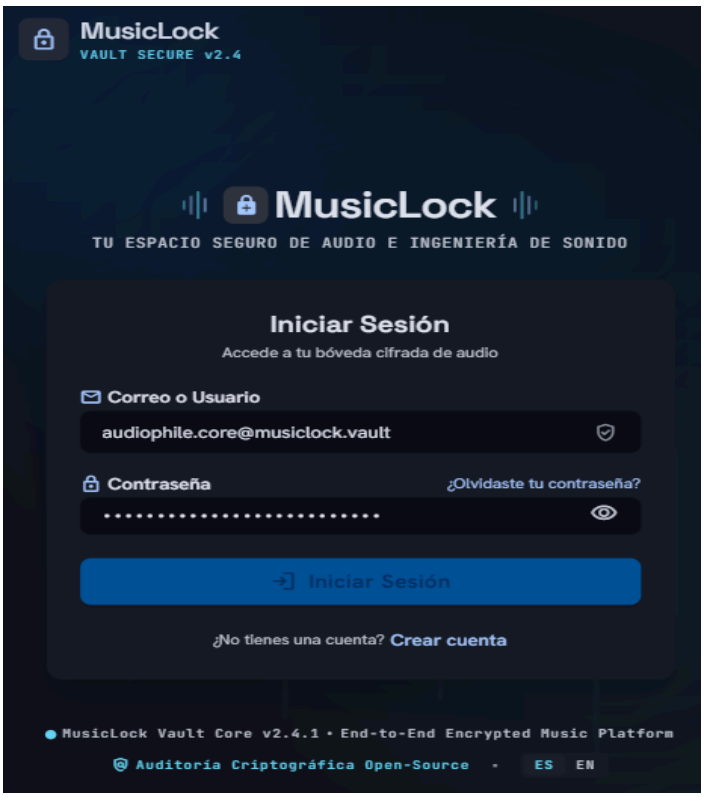


Figure 7. Login Screen

5.3. Workspace

Global navigation panel persistent across authenticated environments.

Sidebar

Provides persistent, high-level navigation across all authenticated sections of the application. It allows users to switch fluidly between the main dashboard, private directories, playlists, and analytical panels, displaying the system's current encryption state at all times.

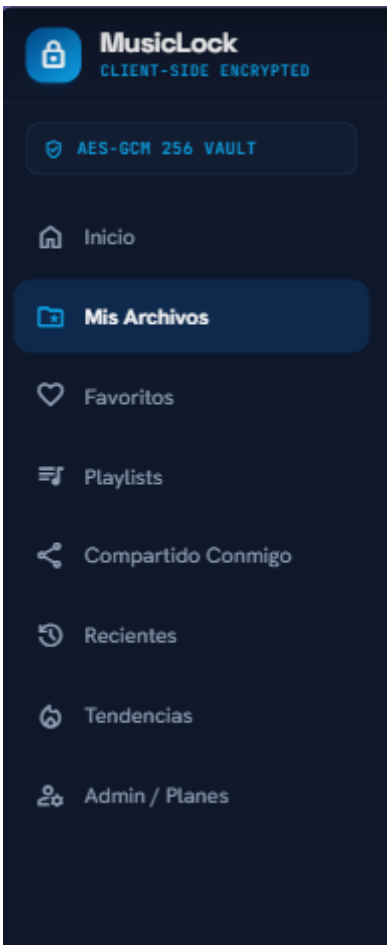


Figure 8. Sidebar menu

Page "Inicio"

High-level summary of active vault space usage, immediate track playback options, and community analytics.

Key Sections:

- Primary action buttons for immediate file upload or new directory creation.
- Recent files tracking spreadsheet detailing track titles, privacy flags, format metrics, and file sizes.
- Data statistics panels recording local streaming loads and overall memory capacity allocation tracking.

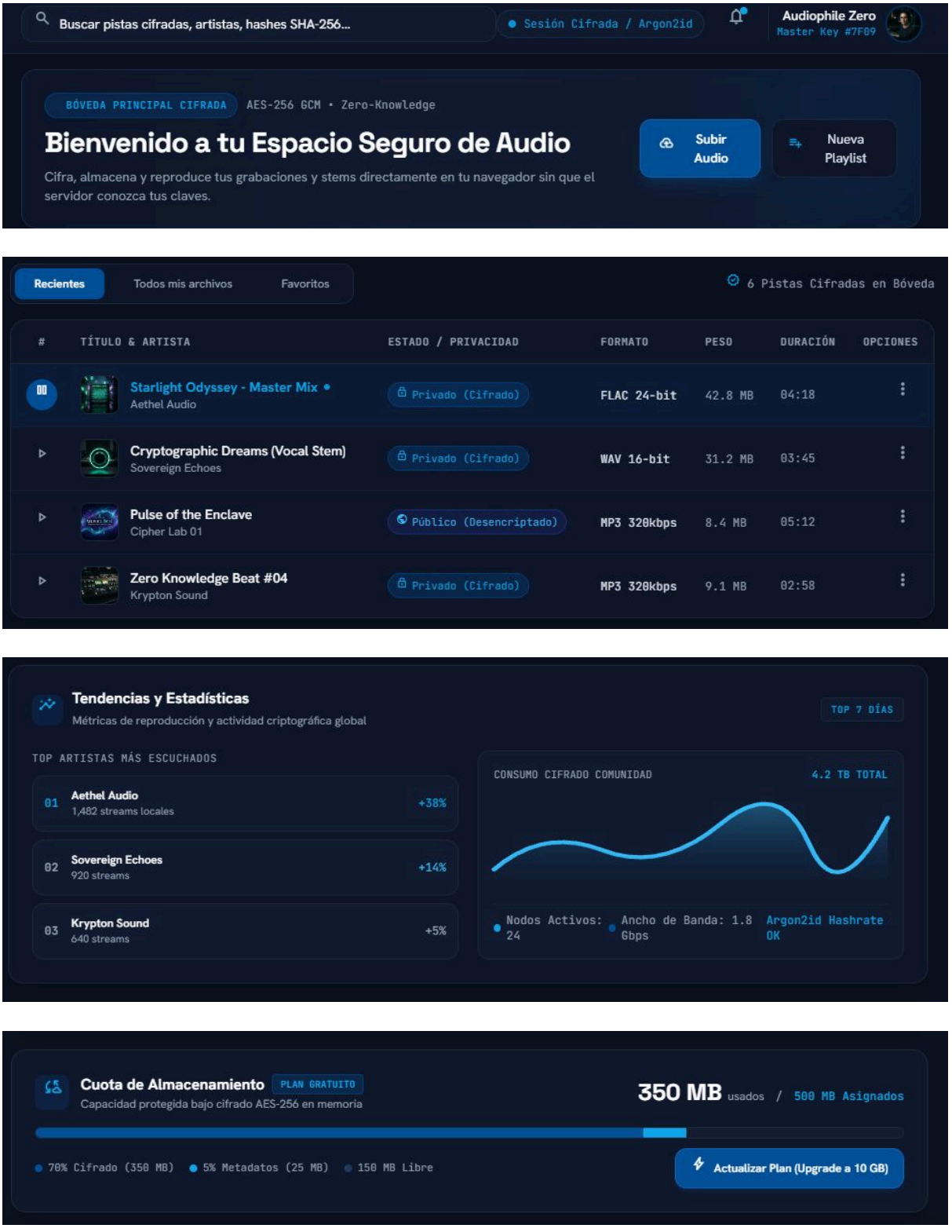


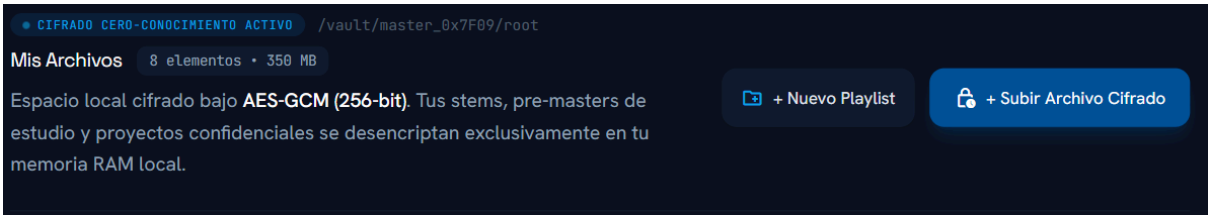
Figure 9. Page-Inicio

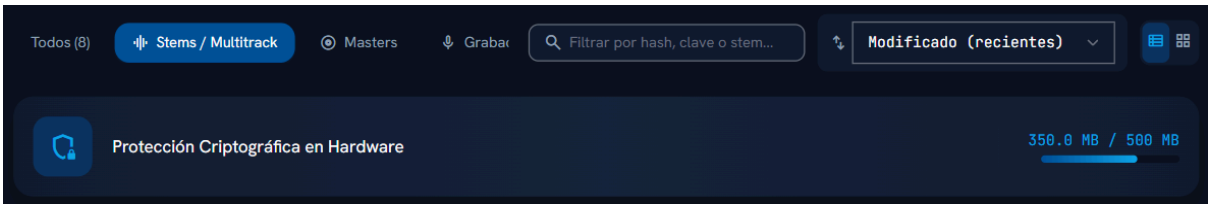
Page "Mis Archivos"

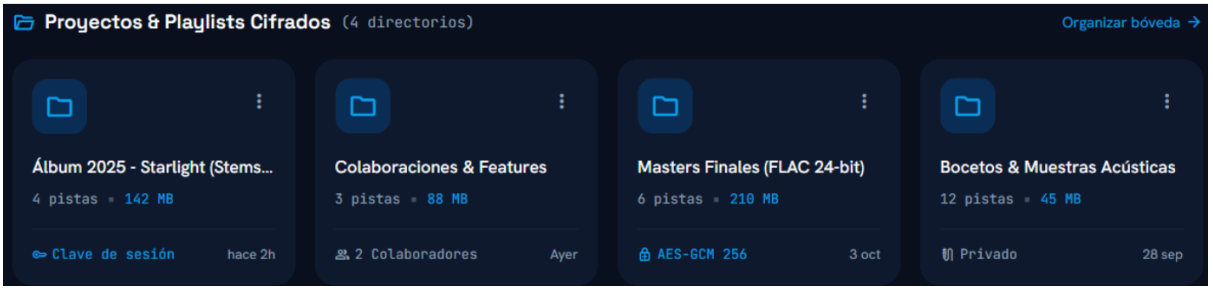
granular management layout of the local user directory.

Key Sections:

- **Vault Status Overview:** Displays real-time metrics of the active hardware-encrypted vault, including technical cryptographic parameters, total storage consumption, and upload actions.
- **Granular Directory View:** Provides organized visual partitions for specialized audio classification, separating raw multi-track stems from finalized audio master files.
- **Secure Playlist Management:** Visualizes private user-created playlists structured as custom secure folders for immediate metadata filtering and track sorting.







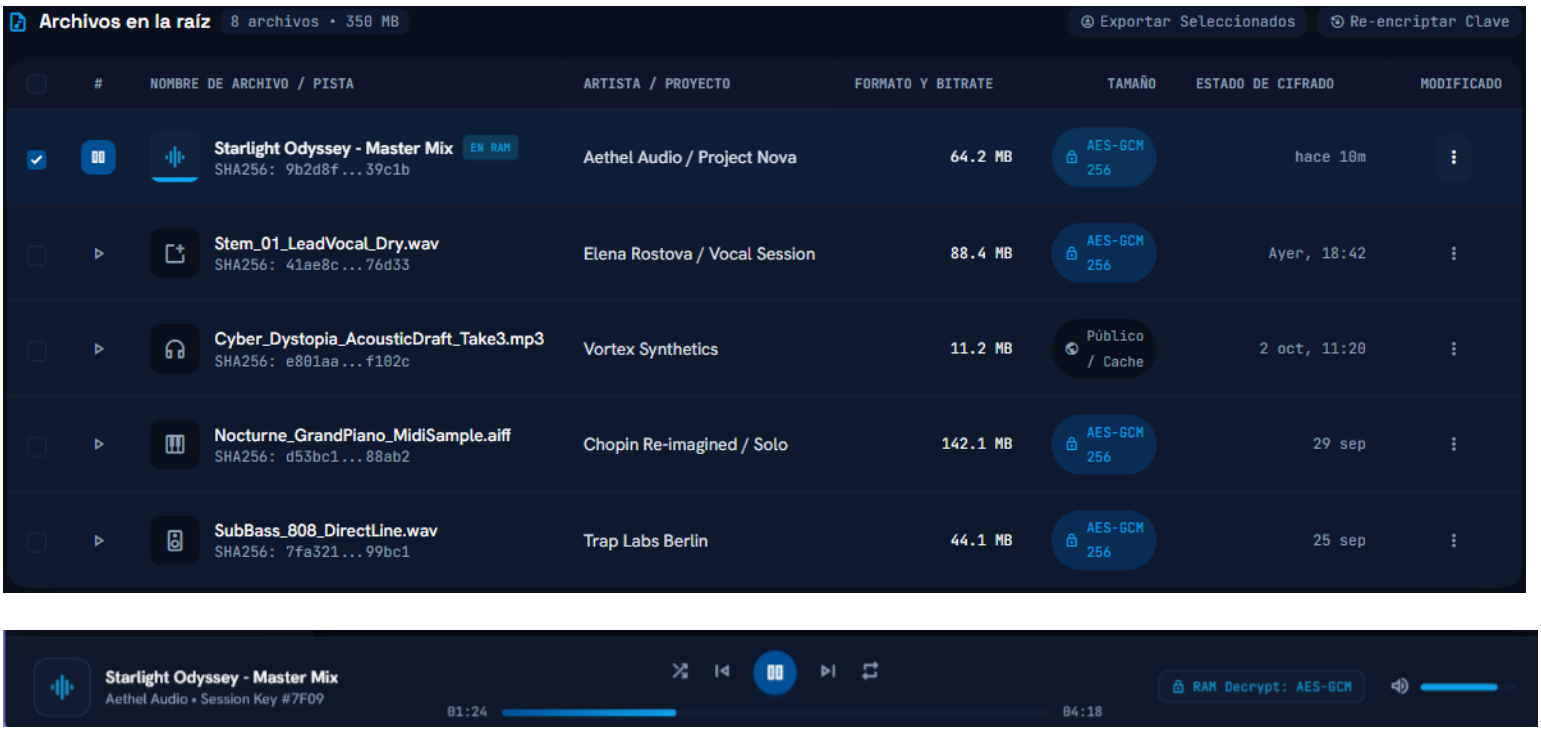


Figure 10.Page -**Mis Archivos**

Page "Favoritos"

A dedicated ecosystem that indexes the user's most critical or high-priority audio tracks. It bypasses full directory structures to grant immediate playback access, streaming directly from cached RAM buffers.

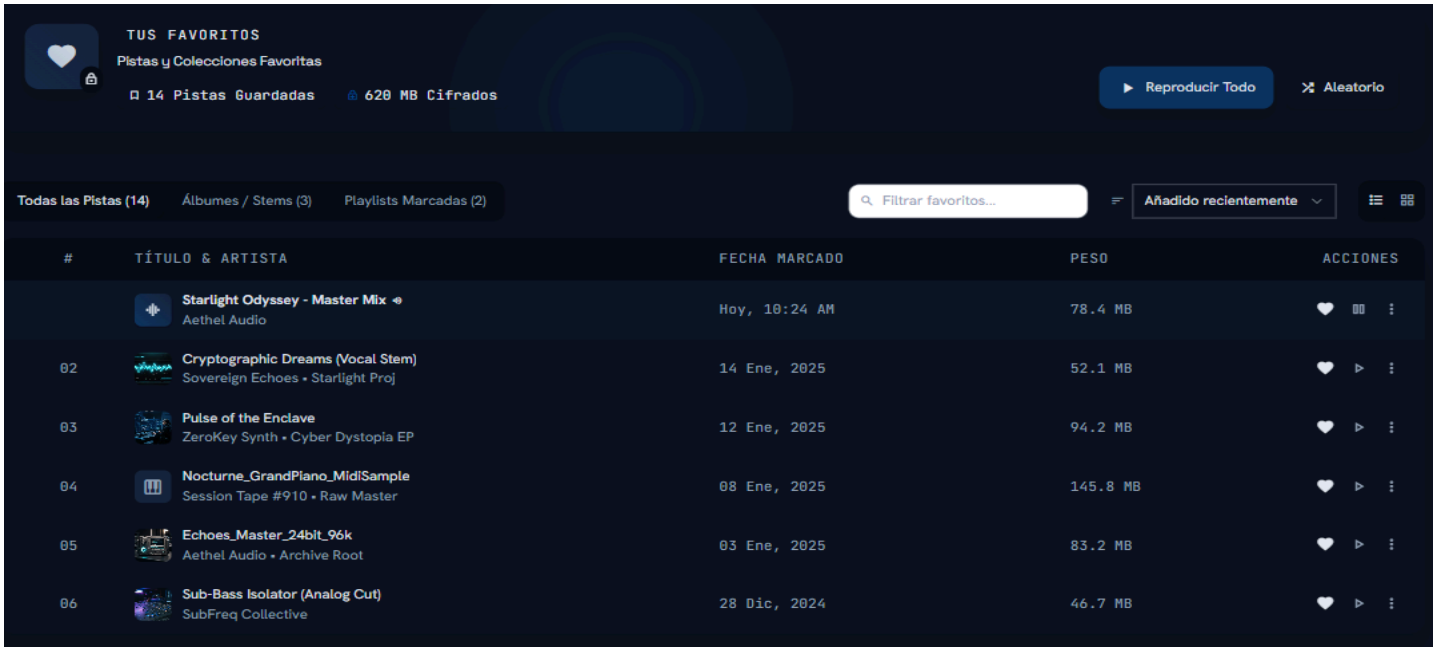


Figure 11. Page -**Favoritos**

Page "PlayLists"

The core organizational workspace where users construct, tag, and categorize independent audio releases or project archives. It supports localized folder generation, metadata filtering, and custom cover image association managed strictly client-side.

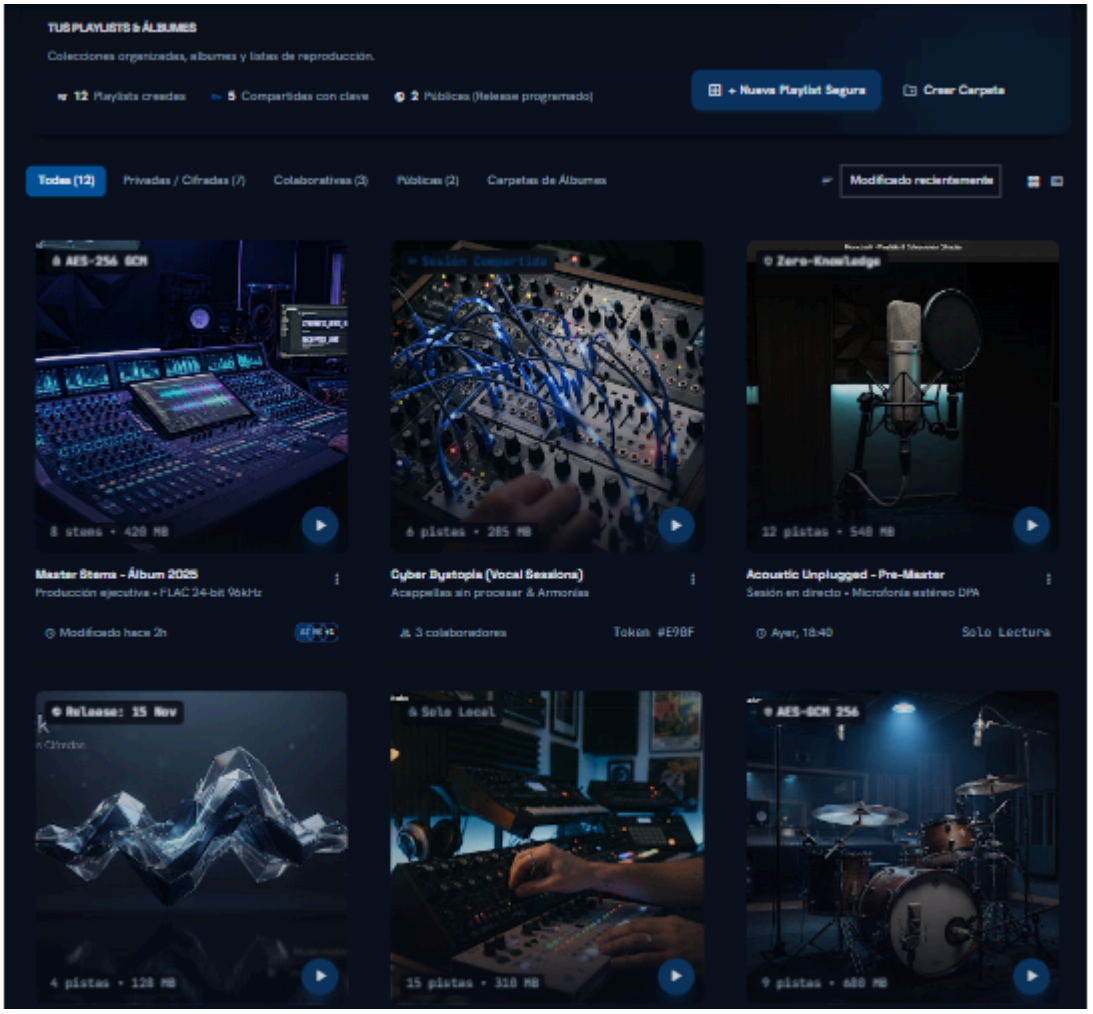




Figure 12. Page -**Play List**

Page "Compartido conmigo"

Manages inbound collaborative streams and external multitrack sharing. This view filters tracks shared by third parties via temporary or permanent cryptographic keys, enabling restricted, secure cross-collaboration without exposing master vault keys.

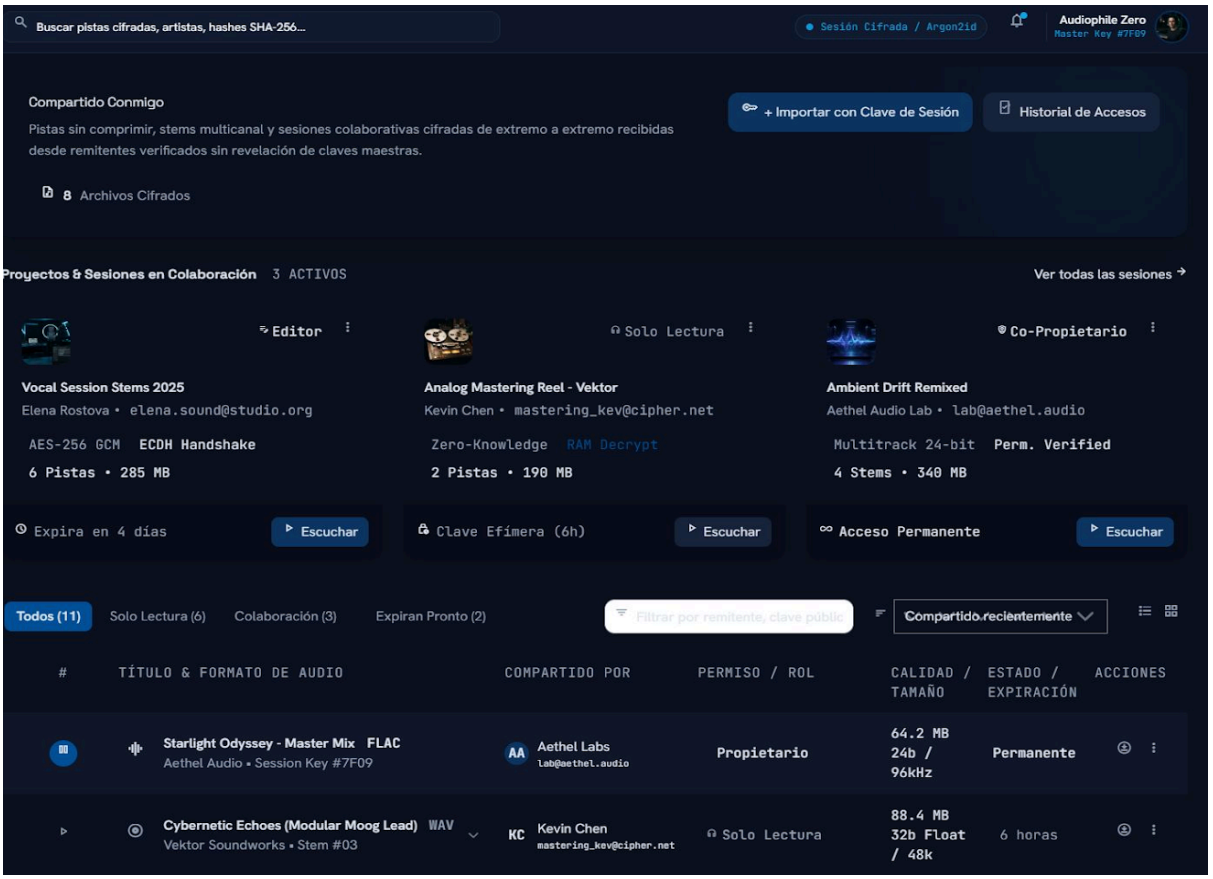


Figure 13. Page -Compartido conmigo

Page "Recientes"

A chronologically sorted log displaying the latest audio files imported, modified, or streamed within the platform. It functions as an optimized shortcut panel for active, ongoing studio production sessions.

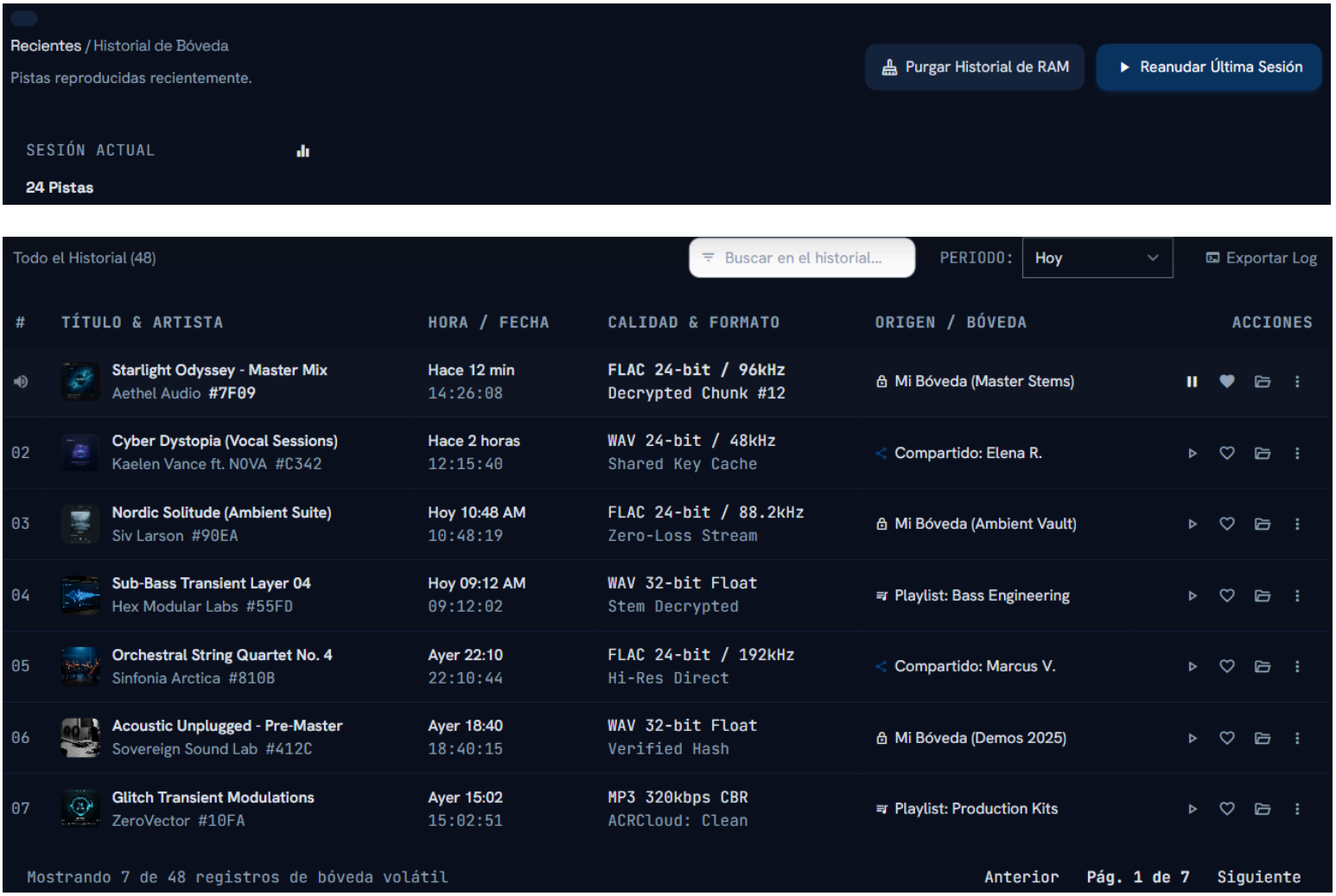


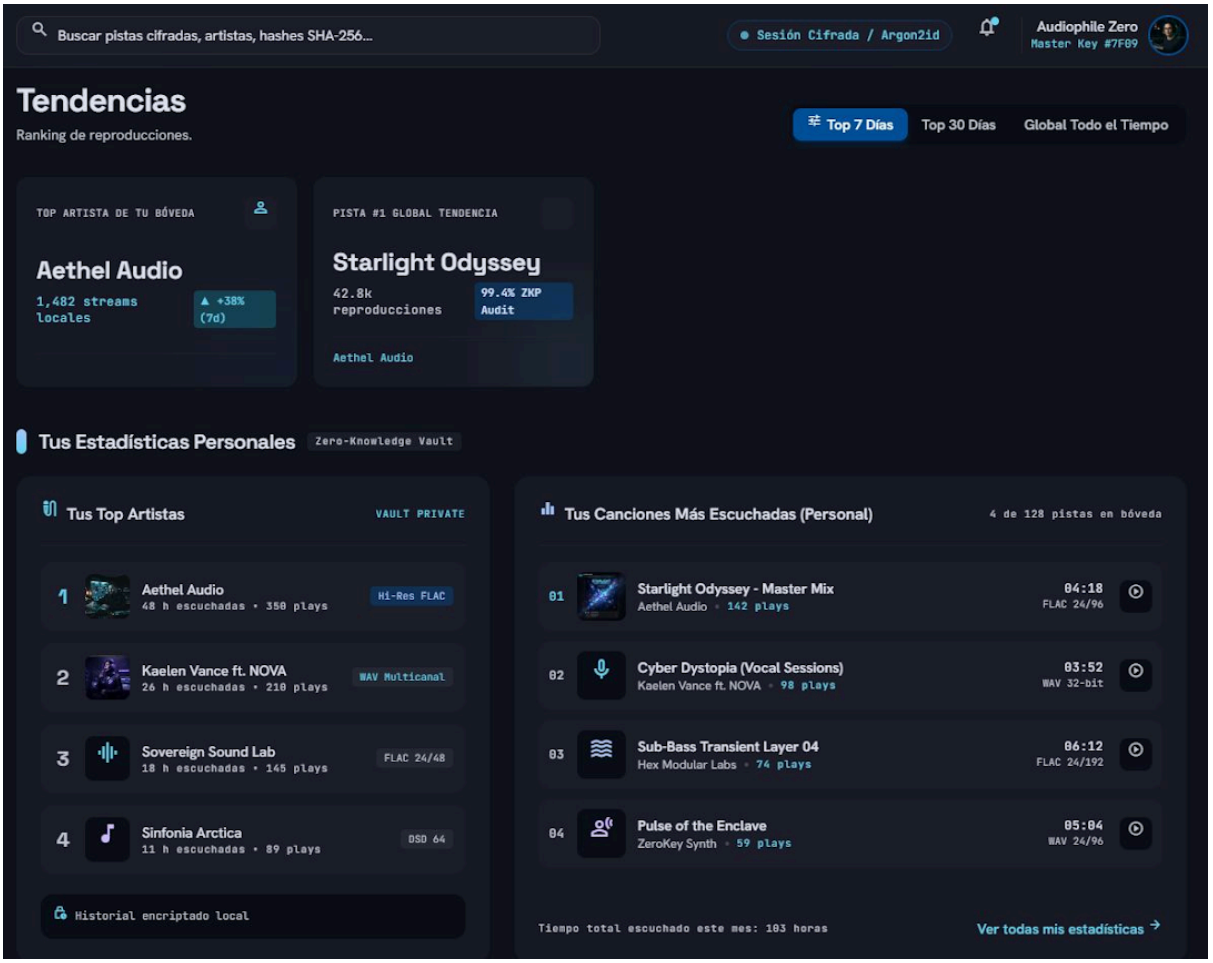
Figure 14. Page -Recientes

Page "Tendencias"

An optimized, privacy-first analytical dashboard that securely aggregates the user's local listening habits and platform-wide interactions. To maintain strict zero-knowledge security, all streaming metrics, top track counts, and listener analytics are processed through decentralized cryptographic audits, protecting intellectual property and user privacy while delivering accurate playback statistics.

Key Sections:

- **Personal Auditing Panel (Tus Estadísticas Personales):** Displays a localized chronological summary of the user's most replayed master tracks and hours spent streaming from their secure vault memory.
- **Encrypted Global Vectors (Global Tendencias):** Visualizes cryptographic public metrics of the platform's overall popular tracks without compromising individual user identity or data tracking logs.
- **Cryptomining / Network Hashrate Feed:** Monitors active nodes, node bandwidth, and active encryption hashrates executing real-time integrity verification across the platform's secure mesh network.



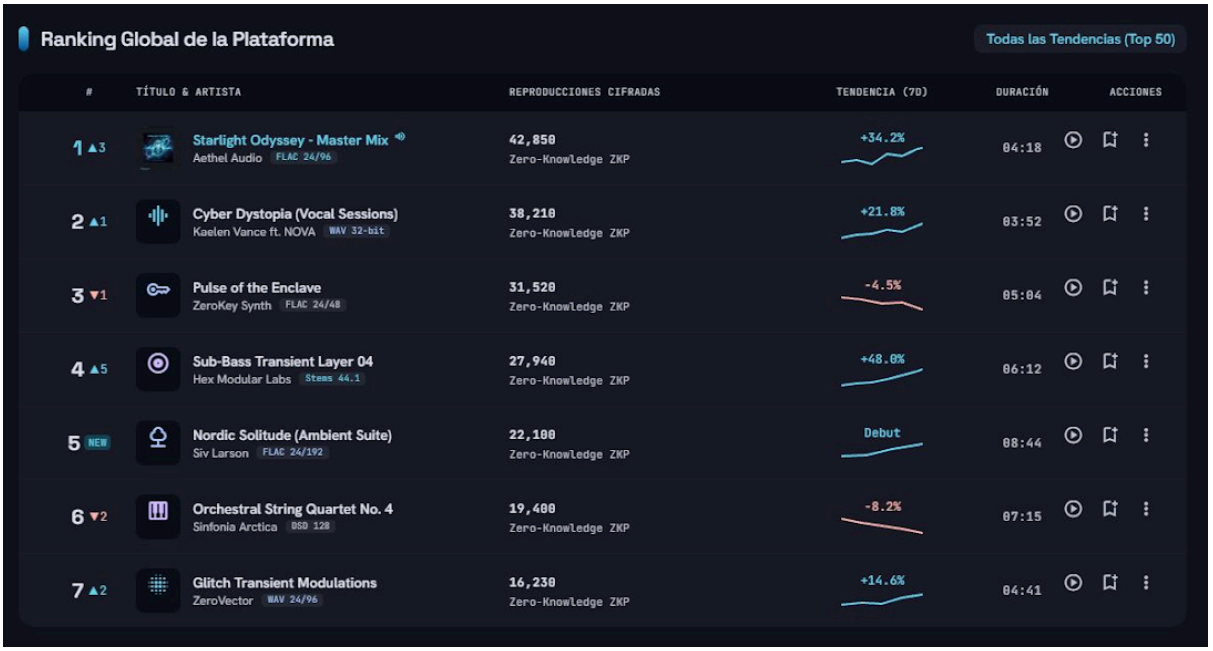


Figure 15. Page -Tendencias

Page "Admin/Planes"

The central administrative panel for account tier maintenance, secure storage allocation diagnostic data, and root privilege verification. It breaks down encryption configurations, current billing workflows, and offers un-tracked payment gateways for zero-knowledge subscription scaling.

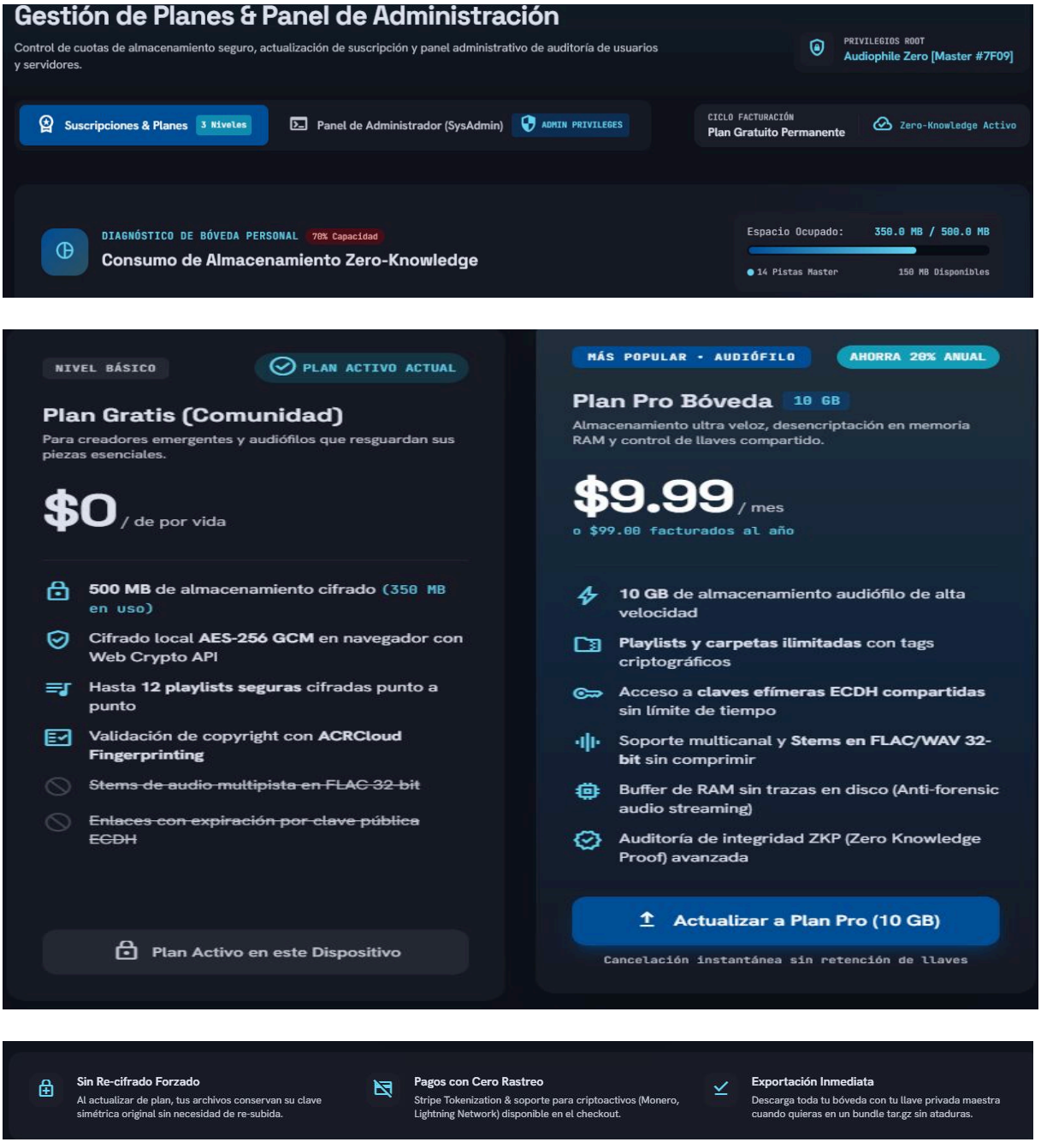


Figure 16. Page -Admin/Planes

5.4. Link of our Mockups Project :Stitch - Projects

Responsible AI Usage Statement

Artificial intelligence tools were utilized in a responsible and complementary manner to support translation, refine technical phrasing, design mockups, and standardize the structure of this document. All CRC cards, domain statements, and technical constraints were defined, validated, and reviewed by human contributors to guarantee factual accuracy, contextual alignment, and system integrity. No confidential data, private credentials, or proprietary assets were shared or processed during these assistance workflows.

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